Bharat Kumar Pathivada

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EDUCATION

- **Ph.D.**, Transportation Systems Engineering, Indian Institute of Technology Bombay, India, 2021
- **M.Tech.**, Transportation Systems Engineering, Indian Institute of Technology Bombay, India, 2021
- **B.Tech.**, Civil Engineering, Jawaharlal Nehru Technological University Kakinada, India, 2012
- **Research Exchange Program**, Hokkaido University, Japan (Summer 2019)
  Team Leader, International Research Skills Program for Sustainable Transportation System and Infrastructure (STSI)

RESEARCH INTERESTS

- Traffic Safety Analysis and Crash Modeling
- Traffic Operations and Simulation
- Human Factors and Behavior Modeling
- Intelligent Transportation Systems

EXPERIENCE

Research Scientist IV [Jan 2026 – Present]
Research Scientist III [Mar 2025 – Dec 2025]
Research Professional II [Jun 2024 – Feb 2025]
Center for Applied Transportation Sciences (CATS)
Department of Civil & Architectural Engineering & Mechanics, University of Arizona

Postdoctoral Research Associate [Aug 2022 – May 2024]
Center for Transportation Research (CTR)
School of Engineering & Applied Sciences, Western Kentucky University

Project Research Scientist [May 2021 – Jun 2022]
Research Associate [Jan 2021 – Apr 2022]
Teaching Assistant [Jul 2014 – Dec 2020]
Department of Civil Engineering, Indian Institute of Technology Bombay

Project Civil Engineer, Punj Lloyd Ltd. [Jun 2012 – Sept 2013]

HONORS AND AWARDS

- All India Rank 706 in the Graduate Aptitude Test in Engineering (GATE) 2014, with a 99.22 percentile.
- Received the Japan Student Service Organization (JASSO) scholarship for short-term study in Japan.
- Received Travel Grant from the Council of Scientific & Industrial Research (CSIR) to attend the 98th Transportation Research Board Annual Meeting, Washington, D.C., USA.
- Best Technical Paper Presentation Award at the National-Level Technical Symposium hosted by Jawaharlal Nehru Technological University Anantapur, India.

SPONSORED RESEARCH PROJECTS

As Principal Investigator

1. Arizona Traffic Survival School Program Efficacy Study. (Co-PI: Dr. Adriana Cimetta, Dr. Yao-Jan Wu). Funding Agency: *Arizona Chapter National Safety COuncil (ACNSC)* ($175,573). [Oct 2026 – Sep 2027]
2. Safety Assessment of Automated Vehicles in the City of Phoenix. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *City of Phoenix, AZ* ($85,000). [Aug 2026 – Jan 2027]
3. Optimizing Traffic Signals Using Multi-Source Data - Phase 10. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *City of Tucson, AZ* ($150,000). [Jul 2026 – Aug 2027]
4. Traffic Engineering Tech Support for the City of Yuma - Phase 4. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *City of Yuma, AZ* ($62,196). [Jul 2026 – Jun 2027]
5. Development of Crash Modification Factors (CMFs) for Signal Modifications and Modernizations. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *City of Phoenix, AZ* ($186,094). [May 2026 – Nov 2027]
6. Safety Evaluation of S. 99th Ave & W. Lower Buckeye Multilane Roundabout in Phoenix, Arizona. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *City of Phoenix, AZ* ($24,925). [Jan 2026 – Jun 2026]
7. Technical Support for Effective Arterial Traffic Management - Phase 2. (Co-PI: Dr. Yao-Jan Wu). Funding Agency: *Town of Marana, AZ* ($80,000). [Aug 2025 – Aug 2026]
8. Traffic Engineering Tech Support for the City of Yuma - Phase 3. (Co-PIs: Dr. Yao-Jan Wu and Dr. Ali Shamshiripour). Funding Agency: *City of Yuma, AZ* ($61,073). [Jul 2025 – Jun 2026]

As Co-Principal Investigator

1. Technical Support for Traffic Operations in Phoenix. (PI: Dr. Yao Jan-Wu). Funding Agency: *City of Phoenix* ($100,000). [July 2026 – Sep 2027]
2. Technical Assistance for the Midtown Central City Transportation Study. (PI: Dr. Yao Jan-Wu). Funding Agency: *City of Phoenix* ($29,123). [Apr 2026 – Jan 2027]
3. Development of Integrated Traffic Data Dashboard for Peoria. (PI: Dr. Yao Jan-Wu). Funding Agency: *City of Peoria* ($125,000). [Jan 2026 – Jan 2027]
4. Development of Arizona-Specific Crash Modification Factors (SPR-789). (PI: Dr. Brendan Russo, Co-PIs: Dr. Steven Gehrke, and Dr. Edward Smaglik). Funding Agency: *Arizona Department of Transportation* (Total: $200,000; UA portion: $50,000). [Aug 2025 – Jun 2027]
5. Optimizing Traffic Signals Using Multi-Source Data - Phase 9. (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Tucson, AZ* ($125,000). [Jul 2025 – Aug 2026]
6. Improving Traffic Operations Using Advanced Methods. (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Casa Grande, AZ* ($40,000). [Feb 2025 – Aug 2025]
7. Engineering Training for Traffic Signal Timing Optimization - Phase 3. (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Phoenix, AZ* ($66,807). [Feb 2025 – Jan 2026]

As Researcher

1. Scalable AI-Driven Tool for Comprehensive Data Collection, Analysis, and Decision Support in Complete Streets. (PI: Dr. Wei Sun, AIWaysion, Co-PI: Dr. Yao Jan-Wu). Funding Agency: *USDOT* (Total: $200,000; UA portion: $50,000). [Nov 2024 – Apr 2025]
2. NCHRP 07-34: Artificial Intelligence for Transportation Systems Management and Operations Applications. (PI: Dr. Yao-Jan Wu). Funding Agency: *National Academy of Sciences* ($450,000). [Jun 2024 – May 2024]

3. Optimizing Traffic Signals Using Multi-Source Data - Phase 8. (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Tucson, AZ* ($125,000). [Jul 2024 – Aug 2025]
4. Traffic Engineering Tech Support for the City of Yuma - Phase 2. (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Yuma, AZ* ($60,000). [Jul 2024 – Jul 2025]
5. Investigating Road User's Compliance with Yellow and Clearance Time Intervals for Signal Timing Design (Phase 3). (PI: Dr. Yao-Jan Wu). Funding Agency: *City of Phoenix, AZ* ($694,140). [Aug 2024 – May 2026]
6. Technical Support for Effective Arterial Traffic Management. (PI: Dr. Yao-Jan Wu). Funding Agency: *Town of Marana, AZ* ($80,000). [Aug 2024 – Jul 2025]
7. Comprehensive Investigation of Motorcycle Crashes in Kentucky. (PI: Dr. Kirolos Haleem). Funding Agency: *Kentucky Transportation Cabinet (KYTC)* ($150,588). [Mar 2023 – May 2024]
8. Safety Investigation of Roundabouts and Restricted Crossing U-Turn (RCUT) Intersections in Kentucky. (PI: Dr. Kirolos Haleem). Funding Agency: *Kentucky Transportation Cabinet (KYTC)* ($316,494). [Aug 2023 – May 2024]
9. Phase 2: Impact of Real-Time Weather Information on Traffic Safety in Kentucky. (PI: Dr. Kirolos Haleem). Funding Agency: *KYTC* ($144,552). [Mar 2024 – May 2024]
10. Statewide Investigation of Distracted Driving Crashes in Kentucky. (PI: Dr. Kirolos Haleem). Funding Agency: *KYTC* ($154,000). [Mar 2023 – May 2024]
11. Impact of Real-Time Weather Information on Traffic Safety in Kentucky. (PI: Dr. Kirolos Haleem). Funding Agency: *KYTC* ($137,386). [Aug 2022 – Jun 2023]
12. Evidence-based Policy Reformation and Implementation to Strengthen Motorised Two-Wheeler Users' Safety. (PI: Dr. Vedagiri Perumal). Funding Agency: *Global Road Safety Partnership (GRSP)* (CHF 80,000). [May 2021 – Jun 2022]
13. Monitoring and Evaluation of Road Safety in Mumbai under Bloomberg Initiative for Global Road Safety (BIGRS). (PI: Dr. Vedagiri Perumal). Funding Agency: *John Hopkins International Injury Research Unit*. [Jun 2015 – Jun 2022]
14. Traffic Speed Management and Improving Safety in Mumbai City, India. (PI: Dr. Vedagiri Perumal). Funding Agency: *United Way Mumbai*. [Jan 2019 – Aug 2021]
15. Review of Traffic and Geometric Design of Oberoi Mall Intersection at Goregaon Mulund Link Road (GMLR) - Western Express Highway (WEH). (PI: Dr. K. V. Krishna Rao, Co-PI: Dr. Vedagiri Perumal). Funding Agency: *Municipal Corporation Greater Mumbai*. [Feb – Jun 2021]
16. Safety audit of Katraj - Swargate BRTS corridor, Pune. (PI: Dr. Vedagiri Perumal). Funding Agency: *Pune Municipal Corporation*. [Jan – Mar 2021]
17. Road safety audit on NH50 (Khed - Sinnar) in the state of Maharashtra. (PI: Dr. Vedagiri Perumal). Funding Agency: *National Highway Authority of India (NHAI)*. [Dec 2015 – Mar 2018]
18. Safety audit of Bus Rapid Transit (BRT) system at Pimpri Chinchwad, Pune. (PI: Dr. Vedagiri Perumal). Funding Agency: *Pimpri Chinchwad Municipal Corporation*. [Jan 2018 – Apr 2018]
19. Design stage audit of Chandni Chowk Intersection in Pune. (PI: Dr. Vedagiri Perumal). Funding Agency: *National Highway Authority of India (NHAI)*. [Jan 2017 – Jun 2017]
20. Traffic Impact Assessment of proposed residential development at Lower Parel, Mumbai. (PI: Dr. Vedagiri Perumal). Funding Agency: *Omkar Relators & Developers Pvt. Ltd*. [Mar 2016 – Dec 2016]
21. Safety evaluation of Pune-Solapur Highway (NH9) in the state of Maharashtra. (PI: Dr. Vedagiri Perumal). Funding Agency: *Pune-Solapur Expressway Private Limited*. [Oct 2015 – Jan 2016]

PUBLICATIONS (Google Scholar | ResearchGate)

Journals

1. **Pathivada, B. K.,** Haleem, K., Khan, T., and Banerjee, A. (2026). Investigating Motorcycle Crash Injury Severity along Kentucky's Rural Two-Lane Roads Using Microscopic Detailed Real-Time Weather Data. *Journal of Transportation Engineering, Part A: Systems*, 152(10):04026096. (Online Access).
2. Geffen, G., Khalilabadi, P, J., **Pathivada, B. K.,** and Wu, Y. J. (2026). Leveraging Multisource Data for a Comprehensive Investigation of Split Failure Occurrences. *Journal of Transportation Engineering, Part A: Systems*, 152(6): 04026028. (Online Access).
3. Khalilabadi, P, J., **Pathivada, B. K.,** and Wu, Y. J. (2026). Exploring Factors Contributing to Red-Light Running Crash Injury Severity at Signalized Intersections using Association Rule Mining. *Journal of Transportation Safety & Security*, 18(8), 1125-1155. (Online Access).
4. **Pathivada, B. K.,** Banerjee, A., Haleem, K., and Khan, T. (2025). Developing Safety Performance Functions for Fatal and Severe Motorcycle Crashes at Intersections. *Transportation Research Record: Journal of Transportation Research Board*, 2679 (12), 373-387. (Online Access).
5. Khan, T., Haleem, K., Banerjee, A., and **Pathivada, B. K.** (2025). Injury Severity Analysis along Major Arterial Roads in Kentucky using High-Resolution Weather Data. *Transportation Research Record: Journal of Transportation Research Board*, 2679 (10), 470-486. (Online Access).
6. Pudasaini, P., **Pathivada, B. K.,** and Wu, Y. J. (2025). Modeling Influence Area for Urban Signalized Intersections using Crowdsourced Trajectory Data. *Transportation Research Record: Journal of Transportation Research Board*, 2679(8), 509-523. (Online Access).
7. **Pathivada, B. K.,** Banerjee, A., Haleem, K., Khan, T., and Justice, D. (2025). Modeling motorcycle crash frequency on rural multilane segments in Kentucky. *Accident Analysis & Prevention*, 218, 108085. (Online Access).
8. Choudhari, T. P., **Pathivada, B. K.,** Maji, A., Jayakumar, M., Damani, J., Perumal, V., Velaga, N. R., and Rao, K. V. K. (2025). Helmet Non-Compliance Among Motorized Two-Wheelers: Insights from Public Opinion Surveys. *Case Studies on Transport Policy*, 20, 101439. (Online Access).
9. **Pathivada, B. K.,** Banerjee, A., Haleem, K., and Khan, T. (2025). Safety Performance Functions for Motorcycle Crashes at Interchange Ramp Segments. *Journal of Safety Research*, 93, 44-54. (Online Access).
10. **Pathivada, B. K.,** and Perumal, V. (2025). Evaluating the Effectiveness of Engineering Countermeasures at Yellow Dilemma Zone: A Driving Simulator-based Study. *Transportation Letters*, 17(1), 49-60. (Online Access).
11. Khan, T., **Pathivada, B. K.,** Banerjee, A., and Haleem, K. (2025). Development of Roundabout Crash-Specific Safety Performance Functions in Kentucky Using Different Techniques. *Journal of Safety Research*, 93, 282-291. (Online Access).
12. Banerjee, A., **Pathivada, B. K.,** Haleem, K., Khan, T., and Justice, D. (2025). Developing Distraction-Related Safety Performance Functions at Interchange Ramp Terminals in Kentucky. *Journal of Safety Research*, 93, 66-78. (Online Access).
13. **Pathivada, B. K.,** Banerjee, A., and Haleem, K. (2024). Investigating motorcycle crashes pre-and post-COVID-19 pandemic along Kentucky's urban roadway segments. *Journal of Safety Research*, 91, 245-257. (Online Access).
14. **Pathivada, B. K.,** Haleem, K., and Banerjee, A. (2024). Investigating the Effect of Microscopic Real-Time Weather Data on Commercial Motor Vehicle Crash Injury Severity in Kentucky. *Transportation Research Record: Journal of Transportation Research Board*, 2678(12), 1659-1676. (Online Access).

15. **Pathivada, B. K.,** Banerjee, A., and Haleem, K. (2024). Impact of Real-Time Weather Conditions on Crash Injury Severity in Kentucky using the Correlated Random Parameters Logit Model with Heterogeneity in Means. *Accident Analysis & Prevention*, 196, 107453.(Online Access).
16. Banerjee, A., Pemberton, W., Woodard, T., McElroy, **Pathivada, B. K.,** and Haleem, K. (2024). Investigating Distracted Driving Crashes in Kentucky Pre- and Post-COVID-19 Pandemic. *Journal of Transportation Safety & Security*, 1-18. (Online Access).
17. Banerjee, A., **Pathivada, B. K.,** Haleem, K., Justice, D., Brittenham, E., and Oliver, J. (2025). Developing Safety Performance Functions for Severe Distraction-Related Crashes along Kentucky's Rural and Urban Two-Lane Roadways. *Transportation Research Record: Journal of Transportation Research Board*, 2679(2), 649-668. (Online Access).
18. Khaddar, S., **Pathivada, B. K.,** and Perumal, V. (2023). Modeling Vehicles Over Speeding Behavior Using a Random Parameter Negative Binomial Approach: A Case Study of Mumbai, India. *Transportation Research Interdisciplinary Perspectives*, 18, 100790. (Online Access).
19. **Pathivada, B. K.,** and Perumal, V. (2022). Investigating Dilemma Zone Boundaries for Mixed Traffic Conditions Using Support Vector Machines. *Transportation Letters*, 14(4), 378-384. (Online Access).
20. Singh, K. M., **Pathivada, B. K.,** Rao, K. R., and Perumal, V. (2022). Driver Behaviour Modelling of Vehicles at Signalized Intersection with Heterogeneous Traffic. *IATSS Research*, 46(2), 236-246. (Online Access).
21. **Pathivada, B. K.,** and Perumal, V. (2019). Analysing Dilemma Driver Behaviour at Signalized Intersection under Mixed Traffic Conditions. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 111-120. (Online Access).

Conference Proceedings and Presentations

1. Moin, M. R. M., Pudasaini, P., **Pathivada, B. K.,** and Wu, Y. J. (2027). Evaluating Sample Size Requirements for Arterial Signal Coordination Performance Using Vehicle Trajectories. *Presented at the 106th Transportation Research Board 2027 Annual Meeting*, Washington, D.C., USA. (TRBAM-27-05153).
2. Adhikhari, A., Shafik, A., **Pathivada, B. K.,** and Wu, Y. J. (2027). Evaluating Pedestrian Hybrid Beacon Coordination Strategies on Urban Arterials: Demand-Dependent Effects on Pedestrian Service and Vehicle Progression. *Presented at the 106th Transportation Research Board 2027 Annual Meeting*, Washington, D.C., USA. (TRBAM-27-04771).
3. **Pathivada, B. K.,** and Shamshiripour, A. (2026). Advancing Transportation Systems in Arizona: The Role of Applied Research and Collaborative Innovation. *Presented at the Roads and Streets Conference*, Oro Valley, AZ. September 24, 2025.
4. Sun, W., Karim, M. M., Hsu, H-M., **Pathivada, B. K.** (2026). Advancing Traffic Safety through Edge AI and Smart Infrastructure: Insights from Urban and Rural Field Deployments. *Presented at the 21st ASCE International Conference on Transportation & Development 2026*, Detroit, Michigan, USA.
5. Banerjee, A., Haleem, K., Khan, T., and **Pathivada, B. K.** (2026). Safety Evaluation of Retroreflective Backplate and Left-Turn Flashing Yellow Arrow at Kentucky's Signalized Intersections. *Presentated at the 21st ASCE International Conference on Transportation & Development 2026*, Detroit, Michigan, USA. (Online Access).
6. Adhikhari, A., Shafik, A., **Pathivada, B. K.,** and Wu, Y. J. (2026). Performance Analysis of Pedestrian Hybrid Beacon Coordination Strategies along Urban Corridors. *Presentated at the 21st ASCE International Conference on Transportation & Development 2026*, Detroit, Michigan, USA.

7. Alnuaimi, Y., Pudasaini, P., **Pathivada, B. K.,** and Wu, Y. J. (2026). Comparative Analysis of Volume- and Trajectory-Based Strategies for Traffic Signal Retiming and Coordination. *Presented at the 105th Transportation Research Board 2026 Annual Meeting*, Washington, D.C., USA. (TRBAM-26-03779).
8. Khalilabadi, P. J., **Pathivada, B. K.,** Wu, Y. J., and Ramos, S. (2026). Evaluating the Impact of Yellow Change Interval Adjustments on Red-Light Running: Evidence from an Empirical Bayes Before–After Study. *Presented at the 105th Transportation Research Board 2026 Annual Meeting*, Washington, D.C., USA. (TRBAM-26-04378).
9. Moin, M. R. M., Pudasaini, P., **Pathivada, B. K.,** and Wu, Y. J. (2026). Interpretable Machine Learning Approach for Analyzing Injury Severity in Single-Vehicle Large Truck Lane Departure Crashes. *Presented at the 105th Transportation Research Board 2026 Annual Meeting*, Washington, D.C., USA. (TRBAM-26-04361).
10. Khalilabadi, P. J., **Pathivada, B. K.,** and Wu, Y. J. (2026). Uncovering Patterns in Red-Light Running Crash Severity at Signalized Intersections: A Data-Driven Analysis Using Association Rule Mining. *Presented at the 105th Transportation Research Board 2026 Annual Meeting*, Washington, D.C., USA. (TRBAM-26-03658).
11. Khalilabadi, P. J., **Pathivada, B. K.,** and Wu, Y. J. (2026). Investigating Crash Severity Involving Driver Non-Compliance at Unsignalized Intersections: Insights from an Interpretable Machine Learning Framework. *Presented at the 105th Transportation Research Board 2026 Annual Meeting*, Washington, D.C., USA. (TRBAM-26-03639).
12. Wu, Y. J., **Pathivada, B. K.,** and Shamshiripour, A. (2025). Advancements in Transportation Systems: The Role of Applied Research in Arizona. *Presented at the Roads and Streets Conference*, Oro Valley, AZ. September 25, 2025.
13. Khalilabadi, P, J., **Pathivada, B. K.,** and Wu, Y. J. (2025). Assessing the Impact of Extended Yellow Durations on Driver Violations: An Empirical Bayes Approach. *Presented at the 20th ASCE International Conference on Transportation & Development 2025*, Glendale, Arizona, USA.
14. Alnuaimi, Y. N., **Pathivada, B. K.,** and Wu, Y. J. (2025). Comparative Analysis of Daytime and Nighttime Crashes in the City of Yuma, Arizona. *Presented at the 20th ASCE International Conference on Transportation & Development 2025*, Glendale, Arizona, USA.
15. Khalilabadi, P, J., Akinboro, T., **Pathivada, B. K.,** and Wu, Y. J. (2025). Exploring Trends and Patterns of Non-Compliant Crashes at Signalized Intersections in Phoenix, AZ. *Presented at the 20th ASCE International Conference on Transportation & Development 2025*, Glendale, Arizona, USA.
16. Geffen, G., Khalilabadi, P, J., **Pathivada, B. K.,** and Wu, Y. J. (2025). Leveraging Multisource Data for Insights into Split Failure Occurrences. *Presented at the 20th ASCE International Conference on Transportation & Development 2025*, Glendale, Arizona, USA.
17. Pudasaini, P., **Pathivada, B. K.,** and Wu, Y. J. (2025). Modeling Influence Area for Urban Signalized Intersections using Crowdsourced Trajectory Data. *Presented at the 20th ASCE International Conference on Transportation & Development 2025*, Glendale, Arizona, USA.
18. **Pathivada, B. K.,** Banerjee, A., Haleem, K., and Khan, T., (2025). Developing Safety Performance Functions for Fatal and Severe Motorcycle Crashes at Intersections. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-00080)
19. **Pathivada, B. K.,** Buhay, N., Justice, D., Banerjee, A., and Haleem, K. (2025). Impact of COVID-19 Pandemic on Distraction-Related Motorcycle Crashes. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-00102).
20. **Pathivada, B. K.,** Banerjee, A., Haleem, K., Khan, T., and Justice, D. (2025). Motorcycle Safety Performance Functions along Kentucky's Rural Multilane Segments. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-00065).

21. Pudasaini, P., **Pathivada, B. K.,** Wu, Y., J. (2025). Modeling Influence Area for Urban Signalized Intersections using Crowdsourced Trajectory Data. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-04769).
22. Haleem, K., **Pathivada, B. K.,** and Banerjee. (2025). Crash Injury Severity Investigation along Interstate-64 in Kentucky. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-00049).
23. Khan, T., Haleem, K., Banerjee, A., and **Pathivada, B. K.** (2025). Injury Severity Analysis along Major Arterial Roads in Kentucky Using High-Resolution Weather Data. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-03891).
24. Banerjee, A., Khan, T., Haleem, K., and **Pathivada, B. K.** (2025). Developing Fatal-and-Injury Distraction-Specific Safety Performance Functions along Ramp Segments. *Presented at the 104th Transportation Research Board 2025 Annual Meeting*, Washington, D.C., USA. (TRBAM-25-00750).
25. **Pathivada, B. K.,** Banerjee, A., Haleem, K., Khan, T., and Justice, D. (2024). Modeling Motorcycle Crash Frequency Along Rural Multilane Segments in Kentucky. *Presented at 9th Roads Safety and Simulation Conference 2024*, Lexington, Kentucky, USA.
26. Khan, T., **Pathivada, B. K.,** Haleem, K., and Banerjee, A. (2024). Assessing the Safety Effectiveness of Roundabouts in the State of Kentucky. *Presented at the 9th Roads Safety and Simulation Conference 2024*, Lexington, Kentucky, USA. (Online Access).
27. Banerjee, A., Pemberton, W., Woodard, T., McElroy, B., **Pathivada, B. K.,** and Haleem, K. (2024). Investigating Distracted Driving Crashes in Kentucky Pre-and Post-COVID-19 Pandemic. *Presented at the 9th Roads Safety and Simulation Conference 2024*, Lexington, Kentucky, USA.
28. Banerjee, A., Woodard, T., **Pathivada, B. K.,** Justice, D., and Haleem, K. (2024). Impact of Retroreflective Backplates and Flashing Yellow Arrows on Distraction-Related Crashes in Kentucky. *Presented at the 9th Roads Safety and Simulation Conference 2024*, Lexington, Kentucky, USA. (Online Access).
29. Banerjee, A., Pemberton, W., Woodard, T., McElroy, B., **Pathivada, B. K.,** and Haleem, K. (2024). Impact of COVID-19 on Severe Distraction-Related Crashes: Comparative Study Pre- and Post-Pandemic. *Presented at the 19th ASCE International Conference on Transportation & Development 2024*, Atlanta, Georgia, USA.
30. Banerjee, A., Woodard, T., **Pathivada, B. K.,** Justice, D., and Haleem, K. (2024). Impact of Retroreflective Backplates and Flashing Yellow Arrows on Distraction Crashes in Kentucky. *Presented at the 19th ASCE International Conference on Transportation & Development 2024*, Atlanta, Georgia, USA.
31. **Pathivada, B. K.,** Buhay, N., Justice, D., Banerjee, A., and Haleem, K. (2024). COVID-19 Pandemic and Distraction-Related Motorcycle Crashes in Kentucky. *Presented at the 19th ASCE International Conference on Transportation & Development 2024*, Atlanta, Georgia, USA. (Online Access).
32. **Pathivada, B. K.,** Haleem, K., and Banerjee, A. (2024). Impact of Microscopic Real-Time Weather Information on Commercial Motor Vehicle Crash Severity along Interstate-65 in Kentucky. *Presented at the 103rd Transportation Research Board 2024 Annual Meeting, Washington*, D.C., USA. (TRBAM-24-00179)
33. **Pathivada, B. K.,** Banerjee, A. and Haleem, K. (2024). Developing Motorcycle Crash-Specific Safety Performance Functions Along Rural Two-Lane Undivided Road Segments in Kentucky Pre- and Post-COVID-19 Pandemic. *Presented at the 103rd Transportation Research Board 2024 Annual Meeting*, Washington, D.C., USA. (TRBAM-24-00532)
34. Banerjee, A., Pemberton, W., Woodard, T., McElroy, B., **Pathivada, B. K.,** and Haleem, K. (2024). Safety Investigation of Distracted Driving Crashes in Kentucky Pre- and Post-COVID-19 Pandemic.

Presented at the 103rd Transportation Research Board 2024 Annual Meeting, Washington, D.C., USA. (TRBAM-24-00205)

35. Banerjee, A., **Pathivada, B. K.,** Justice, D., and Haleem, K. (2024). Comprehensive Investigation of Severe Distraction-Related Crashes Along Kentucky's Rural Two-Lane Roads. *Presented at the 103rd Transportation Research Board 2024 Annual Meeting*, Washington, D.C., USA. (TRBAM-24-01057)
36. **Pathivada, B. K.,** and Perumal, V. (2023). Comparative Analysis of Engineering Countermeasures at Yellow Dilemma Zone: A Driving Simulator Based Study. *Presented at the 102nd Transportation Research Board 2023 Annual Meeting*, Washington, D.C., USA. (TRBAM-23-04551)
37. Choudhari, T. P., **Pathivada, B. K.,** Jayakumar, M., Damani, J., Perumal, V., Maji, A., Velaga, N. R., Rao, K. V. K. (2023). Why Motorized Two-wheeler Users Do Not Wear Helmets? An Investigation Based on Public Opinion Surveys. *Presented at the 16th World Conference in Transport Research (WCTR)*, Toronto, Canada.
38. **Pathivada, B, K.,** Abshire, N., Hartmann, C., Bell, L., Jackson, D., Haleem, K. (2023). Investigating the Impact of Real-Time Weather Variables on Crash Severity along Kentucky's Interstates: Case Studies of I-64, I-65, and I-75. *Presented at the 18th ASCE International Conference on Transportation & Development 2023*, Austin, Texas, USA. (Online Access).
39. **Pathivada, B. K.,** and Perumal, V. (2020). Investigating Indecision Zone Boundaries for Mixed Traffic Conditions: Using Machine Learning Method. *Presented at the 99th Transportation Research Board 2020 Annual Meeting*, Washington, D.C., USA. (TRBAM-20-00813)
40. **Pathivada, B. K.,** and Perumal, V. (2019). Quantifying Dynamic Indecision Zone Boundaries for Heterogeneous Traffic Conditions. *Presented at the 98th Transportation Research Board 2019 Annual Meeting*, Washington, D.C., USA. (TRBAM-19-04084)
41. **Pathivada, B. K.,** and Perumal, V. (2019). Analyzing Stop/Go Behaviour at the Onset of Yellow Signal using Support Vector Machines and Decision Trees. *Presented at the 15th World Conference in Transport Research (WCTR)*, Mumbai, India.
42. **Pathivada, B. K.,** and Perumal, V. (2018). Evaluating the Location of Indecisive Zone at Signalized Intersection using Support Vector Machines. *Presented at the 3rd Recent Advances in Traffic Engineering (RATE)*, Surat, India.
43. **Pathivada, B. K.,** and Perumal, V. (2018). Analyzing Location of Dilemma Zone at Signalized Intersections for Indian Traffic Conditions. *Presented at the 78th IRC Annual Session*, Bengaluru, Karnataka, India.
44. **Pathivada, B. K.,** and Perumal, V. (2018). Modeling Driver Behaviour at Signalized Intersection Dilemma Zone under Mixed Traffic Conditions. *Presented at the 97th Transportation Research Board 2018 Annual Meeting*, Washington, D.C., USA. (TRBAM-18-02754)
45. **Pathivada, B. K.,** and Perumal, V. (2017). Modeling Driver Behaviour in Dilemma Zone under Mixed Traffic Conditions. *Presented at the 20th EURO Working Group on Transportation Meeting (EWGT)*, Budapest, Hungary. (Online Access).
46. **Pathivada, B. K.,** and Perumal, V. (2017). Driver Behavior Models at Dilemma Zone: A Review. *Presented at the 17th COTA International Conference of Transportation Professionals (CICTP)*, Shanghai, China.
47. **Pathivada, B. K.,** and Perumal, V. (2017). Analysis of Driver's Stop/Go Behavior at the Yellow Onset under Mixed Traffic Conditions. *Presented at the International Conference on Advances in Highway Engineering & Transportation (ICAHETS)*, Negombo, Sri Lanka.
48. **Pathivada, B. K.,** and Perumal, V. (2017). Modeling Acceleration Characteristics of Motorized Three-Wheeler. *Presented at the International Conference on Advances in Highway Engineering & Transportation (ICAHETS)*, Negombo, Sri Lanka.

PROFESSIONAL ACTIVITIES

- Member, Institute of Transportation Engineers (ITE) [Aug 2026 – Present]
- Member, Intelligent Transportation Society of Arizona [Oct 2024 – Present]
- Center Manager, Center for Applied Transportation Sciences (CATS) [Nov 2024 – Present]
- Member, TRB Standing Committee on Regional Transportation Systems Management and Operations (ACP10) [Apr 2025 – May 2025]

Reviewer for Journals

- Transportation Research Part F: Traffic Psychology and Behaviour (Elsevier)
- Safety Science (Elsevier)
- Transportmetrica A: Transport Science (Taylor & Francis)
- Transportmetrica B: Transport Dynamics (Taylor & Francis)
- Travel Behaviour and Society (Elsevier)
- Scientific Reports (Nature Portfolio, Springer Nature)
- Transportation in Developing Economies (Springer)

Reviewer for Conferences

- Transportation Research Board (TRB) Annual Meeting
- International Conference on Transportation Planning and Implementation Methodologies for Developing Countries (TPMDC)
- Conference of Transportation Research Group of India (CTRG)
- COTA (Chinese Overseas Transportation Association) International Conference of Transportation Professionals (CICTP)
- Conference on Transportation Systems Engineering and Management (CTSEM)
- Transportation Infrastructure Projects: Conception to Execution (TIPCE)

Other Professional Services

- Sponsorship & Finance Committee Coordinator, 15th World Conference on Transport Research (WCTR), May 2019, Indian Institute of Technology Bombay.
- Finance Committee Coordinator, 4th Conference of Transportation Research Group of India (CTRG), December 2017, Indian Institute of Technology Bombay.
- Student organizing member, five-day technical workshop on Road Safety Audit for Public Works Department Maharashtra, February 2017, Indian Institute of Technology Bombay.
- Student organizing member, Review workshop of Indian Highway Capacity Manual (RIHCAM), March 2016, Indian Institute of Technology Bombay.
- Student organizing member, 12th International Conference on Transportation Planning & Implementation Methodologies for Developing Countries (TPMDC), December 2016, Indian Institute of Technology Bombay.
- Mentor, Hackathon 2016 'Make in India initiative' at Indian Institute of Technology Bombay.
- Student organizing member, 11th TPMDC, December 2014, Indian Institute of Technology Bombay.

Invited Talks and Panel Discussions

- Panelist, AI in Transportation Now, ITS Arizona, 2025.
- Invited Speaker, Effective Enforcement of Two-Wheeler Safety Provisions: Issues, Challenges and Way Forward. Consultation with Enforcement Officials on MVA Act, 2019: Challenges and Way Forward, India, 2021.